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Construction of DNA codes by using algebraic number theory



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ABSTRACT

The canonical structure of DNA has four bases – Thymine (T), Adenine (A), Cytosine (C), and Guanine (G) – and DNA codes are regarded as words over the alphabet set $\Sigma = \{A, C, G, T\}$, satisfying certain combinatorial conditions. Good DNA codes are desirable for DNA computation, DNA microarray technologies and molecular barcodes, etc. One of the main tasks in DNA code designing is to build more codewords and better GC -content for given fixed word length n . Existing heuristic methods work well for small n . In this paper, we present a systematic method for constructing good DNA codes for large n by using irreducible cyclic codes. Being different from traditional DNA constructions, our method is based on algebraic number theory rather than classical heuristic algorithms and the conventional coding theory. Furthermore, comparing with the traditional DNA codes, our codes have larger number of codewords and better GC -content. As far as we know, it is the very first time to

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utilize irreducible cyclic codes for constructing a type of DNA codes.

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1. Introduction

DNA codes are the sets of words of fixed length over the genetic alphabet $\{A, C, G, T\}$. Meanwhile, hybridizations of DNA follow the principle that short single strands of DNA should be hybridized with their Watson–Crick complements, i.e., $\overline{A} = T$, $\overline{T} = A$, $\overline{C} = G$, and $\overline{G} = C$.

DNA codes have wide applications for storage and retrieval of information in synthetic DNA strands. DNA codes are also used for DNA computation [1,3,7–10,12,20,22], as probes in DNA microarray technologies [24] and as molecular barcodes for chemical libraries [5]. In these applications, the primary property required is that short single strands of DNA should be hybridized with their Watson–Crick complements, but other undesirable hybridizations should be unlikely [3]. This is achieved by specifying a minimum Hamming distance between words and as well as between the reverse complements of words.

Another desirable property concerns with the melting temperature. DNA melting is the process by which double-stranded DNA unwinds and separates into single-stranded strands through the breakage of hydrogen bonding between the bases. Similar melting temperatures could be achieved approximately by ensuring that each word contains the same number of positions, which are either G or C.

By employing the notation from code theory, we use $\mathcal{D} = (n, M, d, w)_4$ for denoting a DNA code defined over the genetic alphabet $\{A, C, G, T\}$, the meanings of corresponding parameters are specified as follows: n is the length of the code; M is the number of the codewords; d is the minimum Hamming distance¹; w is GC-content, which is defined as the number of letters of each codeword either as G or C.

The DNA codes designing problem is about to find the largest possible set of DNA codewords, each of length n , satisfying certain combinatorial conditions [23,4,2]. Let $\mathcal{D} = (n, M, d, w)_4$ and $c = (c_0, c_1, c_2, \dots, c_{n-1}) \in \mathcal{D}$, then the conditions are presented as follows:

- (1) (HD) $H(c, c') \geq d, \forall c, c' \in \mathcal{D}, c \neq c'$
- (2) (R) $H(c, c'^{-1}) \geq d, \forall c, c' \in \mathcal{D}$

¹ Note that here, Hamming distance between two codewords x and y , denoted by $H(x, y)$, is defined as the number of positions i at which the i th letter in x differs from the i th letter in y . For instance, $H(CTG, ATGG) = 2$.

- (3) (RC) $H(c, \bar{c}^{-1}) \geq d, \forall c, c' \in \mathcal{D}$
- (4) (GC) $GC\text{-content}$ is constant w for any $c \in \mathcal{D}$,

where $c^{-1} = (c_{n-1}, c_{n-2}, c_{n-3}, \dots, c_0)$ denotes the reverse of c [18], and $\bar{c}^{-1} = (\bar{c}_{n-1}, \bar{c}_{n-2}, \bar{c}_{n-3}, \dots, \bar{c}_0)$ denotes the reverse Watson–Crick complement of c .

Actually, it's rather difficult to design suitable DNA codes satisfying all conditions with large n [20,6,12]. It is also desirable in the applications that the number of codewords should be as large as possible, as this allows the storage of more information [20]. In most of the cases, the maximum number of codewords is obtained when the $GC\text{-content}$ w is $\lfloor \frac{n}{2} \rfloor$ [21,15]. For example, when $n = 4$, $d = 3$, the corresponding DNA codes satisfying conditions (1), (3) and (4) (not unique) are the set $\{TCCT, CAGT, GAAG, TGAC, CCAA, AAC C\}$.

Currently, scientists are eager on constructing the valuable DNA codes satisfying two or three conditions in the real-world applications. Until now, they have tried some effective methods such as classical heuristic algorithms and linear codes in code theory for constructing DNA codes with good properties. In 2005, Philippe Gaborit and Oliver D. King [13] utilized error-correcting codes in code theory and a combination of lexicographic technique with stochastic search for proposing a new construction of DNA codes satisfying conditions (3) and (4) with length $n \leq 20$. In 2008, Derek H. Smith et al. [21] combined four local search algorithms and a variable neighbourhood search approach for constructing valuable DNA codes satisfying conditions (1), (3) and (4) for $n \leq 20$. In 2011, Derek H. Smith et al. [23] designed new DNA codes that satisfy the conditions (1) and (4) for $n \leq 30$ using generator polynomials in cyclic codes. In 2012, Niema Aboluion et al. [4] took an advantage of similar approach for constructing new DNA codes satisfying conditions (1), (3) and (4) for $n \leq 30$. However, the length n of DNA codes is still not large enough ($n \leq 30$). Therefore, it's imminent for designing new DNA codes with valuable properties and having larger length.

In this paper, we adopt methods from algebraic number theory, in particular, irreducible cyclic codes in F_4 , for constructing DNA codes with useful properties. We use the relationship between Gauss period and weight formulas of irreducible cyclic codes [13] for computing the parameters of the corresponding DNA codes. At first, we construct DNA codes that satisfy the conditions (1) and (4). Furthermore, we use a post-computation algorithm for finding suitable parameters for these DNA codes that could satisfy the conditions (1), (3) and (4), simultaneously. Comparing with the traditional DNA codes, a remarkable improvement of our method lies in that our DNA codes have larger number of codewords ($M < 7000$) with larger length ($n < 1000$) and better $GC\text{-content}$ ($w \approx \lfloor \frac{n}{2} \rfloor$). In fact, by using our method, one can get DNA codes, satisfying conditions (1), (3) and (4), for any numbers of n , if there exists a positive integer s such that $e = \frac{4^s - 1}{n}$ is a self-conjugate prime. In other words, our method is legitimate for the construction of DNA codes with adequate larger lengths.

2. Preliminaries

Definition 1 (*Additive character*). (See [14].) Let $q = p^n$ and F_q be a finite field. Then an additive character is a map $\gamma : F_q \rightarrow \mathbb{C}^*$ that satisfies $\gamma(a+b) = \gamma(a)\gamma(b)$ for all $a, b \in F_q$, where $\mathbb{C}^*$ is the nonzero complex number set.

Definition 2 (*Trace*). (See [14].) Let $q = p^n$. Then a trace is a map $T : F_q \rightarrow F_p$ defined by $T(\alpha) = \alpha + \alpha^p + \alpha^{p^2} + \cdots + \alpha^{p^{n-1}}$ for all $\alpha \in F_q$. T could also be denoted by T_p^q .

Generally, F_q can be viewed as a vector space of F_p . Let $v_1, v_2, \dots, v_n$ be a base of F_q . Then $a = a_1v_1 + a_2v_2 + \cdots + a_nv_n$ ($a_k \in F_p$) for all $a \in F_p$. Let $\zeta_p = \exp \frac{2\pi i}{p} \in \mathbb{C}^*$, $\gamma(a) = \zeta_p^j$ ($0 \leq j \leq p-1$), then $\gamma_b : F_q \rightarrow \langle \zeta_p \rangle$ is denoted by $\gamma_b(a) = \zeta_p^{T(ab)}$ for all $a, b \in F_q$.

Definition 3 (*Multiplicative character*). (See [14].) Let $q = p^n$, F_q be a finite field. Then a multiplicative character is a map $\chi : F_q^* \rightarrow \mathbb{C}^*$ that satisfies $\chi(ab) = \chi(a)\chi(b)$ for all $a, b \in F_q^*$, where F_q^* is a cyclic multiplicative group of order $q-1$ with generator α and $\mathbb{C}^*$ is the nonzero complex number set. Let $\zeta_{q-1} = \exp \frac{2\pi i}{q-1} \in \mathbb{C}^*$, then $\chi(\alpha) = \zeta_{q-1}$, $\chi(\alpha^j) = \zeta_{q-1}^j$ ($0 \leq j \leq q-2$).

Definition 4 (*Gauss sum*). (See [19,16].) Let γ and χ be the corresponding additive character and multiplicative character of F_q , respectively. Then Gauss Sum of F_q is defined by $G(\chi, \gamma) = \sum_{x \in F_q^*} \chi(x)\gamma(x)$.

Definition 5 (*Gauss period*). (See [11].) Let $q = 4^s$, $F_q = \langle \alpha \rangle$, e the cyclotomic number, $q-1 = e \cdot n$. $T : F_q \rightarrow F_4$ and $R : F_4 \rightarrow F_2$ are two traces. $\sigma = R \circ T : F_q \rightarrow F_2$ is the composite trace. $F_4^* = \{1, \omega, \omega^2\} = \langle \omega \rangle = \langle \alpha^{\frac{q-1}{3}} \rangle$, $\theta = \alpha^e$, $\langle \theta \rangle$ is a subgroup of F_q^* with order n , the coset representation $C_\lambda = \alpha^\lambda \langle \theta \rangle$ is the e -cyclotomic class, $\chi(\alpha) = \zeta_e$, then Gauss Period is defined by

$$\eta_\lambda = \sum_{y \in C_\lambda} (-1)^{\sigma_2^2(y)} = \frac{1}{e} \sum_{l=0}^{e-1} G(\chi^l) \overline{\zeta_e^{\lambda l}} \quad (1)$$

where $\lambda = 0, 1, 2, \dots, e-1$, $G(\chi) = G(\gamma_1, \chi) = \sum_{x \in F_q^*} G(\gamma_1(x), \chi(x))$ is Gauss sum, $\chi^l = \zeta_{q-1}^{jl}$ ($0 \leq j \leq q-2$).

Definition 6 (*Self-conjugate*). (See [11].) Let $q = p^s$, $q-1 = e \cdot n$, χ be the multiplicative character of F_q with order e . If there exists a positive integer t s.t. $p^t \equiv -1 \pmod{e}$, then e is called self-conjugate.

Corollary 1. (See [11].) When e is self-conjugate, $p^{2t} \equiv 1 \pmod{e}$. $\sigma_p(G(\chi)) = G(\chi^p) = G(\chi) = \sigma_{p^t}(G(\chi)) = G(\overline{\chi}) = \chi(-1)\overline{G(\chi)}$.

Lemma 1. (See [25,11].) Let $q' = q^m$, γ and χ be the corresponding additive character and multiplicative character of F_q , respectively. $\gamma^{(m)}$ and $\chi^{(m)}$ are the corresponding additive character and multiplicative character of $F_{q'}$, respectively. If $\chi \neq \varepsilon$ (the map of constant 1), then Hasse–Davenport lift formula holds:

$$-G(\chi^{(m)}, \gamma^{(m)}) = (-G(\chi, \gamma))^m \quad (2)$$

Theorem 1. Let $p = 2$, $e \geq 3$, $(e, 2) = 1$ and t be the smallest positive integer s.t. $2^t \equiv -1 \pmod{e}$. χ is the multiplicative character of F_q ($q = 2^{2t}$) with order e , $\chi^{(s)}$ is multiplicative character of $F_{q'}$ ($q' = q^{2^{ts}}$, $s \geq 1$) with order e , then:

- (1) Gauss sum $G(\chi) = 2^t$.
- (2) Gauss sum $G(\chi^{(s)}) = 2^{ts}(-1)^{s-1}$.

Proof. (1) When $p = 2$, from Definition 3, we have $\chi(-1) = \chi(1) = 1$. Furthermore, from Corollary 1, $G(\chi) = \sigma_{2^t}(G(\chi)) = G(\overline{\chi}) = \chi(-1)\overline{G(\chi)}$, $G(\chi)^2 = G(\chi)\overline{G(\chi)} = q = 2^{2t}$. Hence, we get $G(\chi) = 2^t$.

(2) From Lemma 1 and the aforementioned conclusion, we have, $-G(\chi^{(s)}) = (-G(\chi, \gamma))^s = (-2^t)^s = 2^{ts}(-1)^s$. Therefore, $G(\chi^{(s)}) = 2^{ts}(-1)^{s-1}$. $\square$

In the case, when e is odd prime and self-conjugate, we only need to compute η_0 and η_1 .

Theorem 2. Let $q = 4^s$, $q - 1 = e \cdot n$, χ be the multiplicative character of F_q ($q = 4^s$, $s \in \mathbb{Z}^+$) with order e . If e is odd prime and self-conjugate, then

- (1) $\eta_0 = \frac{1}{e}(-1 + (e-1)2^s(-1)^{\frac{s}{e}-1})$
- (2) $\eta_1 = \frac{1}{e}(-1 + 2^s(-1)^{\frac{s}{e}})$

Proof. (1) From Definition 5, $\eta_0 = \frac{1}{e} \sum_{j=0}^{e-1} G(\chi^j) \overline{\zeta_e^{\lambda j}} = \frac{1}{e}(G(\varepsilon) + G(\chi) + G(\chi^2) + \cdots + G(\chi^{e-1})) = \frac{1}{e}(-1 + (e-1)2^s(-1)^{\frac{s}{e}-1})$.

(2) Also from Definition 5, $\eta_1 = \frac{1}{e} \sum_{j=0}^{e-1} G(\chi^j) \overline{\zeta_e^{\lambda j}} = \frac{1}{e}(-1 + G(\chi)(\zeta_e + \zeta_e^2 + \cdots + \zeta_e^{e-1})) = \frac{1}{e}(-1 + 2^s(-1)^{\frac{s}{e}})$. $\square$

Definition 7 (Irreducible cyclic codes). (See [17,6].) Let $(n, 2) = 1$, s be the smallest positive integer s.t. $4^s \equiv 1 \pmod{n}$. Let $q = 4^s$, $F_q = \langle \alpha \rangle$, e the cyclotomic number, $\theta = \alpha^e$, $q - 1 = e \cdot n$. $T : F_q \rightarrow F_4$ and $R : F_4 \rightarrow F_2$ are two traces. $\sigma = R \circ T : F_q \rightarrow F_2$ is the composite trace. Then irreducible cyclic codes $\mathbb{D} = \{c_\beta | \beta \in F_q\}$, where $c_\beta = (T_4^q(\beta), T_4^q(\beta\theta), \dots, T_4^q(\beta\theta^{n-1})) \in F_4^n$.

Lemma 2. (See [11].) Let (n, M, d) be the parameter of irreducible cyclic codes $\mathbb{D}$, $M = 4^s$, $N_\beta(a)$ be the number of weight $a \in F_4$, then

$$N_{\beta}(0) = \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) \quad (3)$$

$$N_{\beta}(1) = \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) \quad (4)$$

$$N_{\beta}(\omega) = \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) \quad (5)$$

$$N_{\beta}(\omega^2) = \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) \quad (6)$$

where $\frac{q-1}{3} \equiv l \pmod{e}$, $\beta = \alpha^{\lambda}$, $\lambda = 0, 1, 2, \dots, e-1$.

3. Main results

In fact, according to the algebraic number theory point of view, the genetic alphabet $\{A, C, G, T\}$ could be easily viewed as the elements of finite field F_4 : $A \rightarrow 0$, $C \rightarrow \omega$, $G \rightarrow \omega^2$, $T \rightarrow 1$. The Watson–Crick complements correspond to the complement in F_4 ($\overline{0} = 1$, $\overline{1} = 0$, $\overline{\omega} = \omega^2$, $\overline{\omega^2} = \omega$). Then, DNA codes could be defined in F_4 . Generally speaking, a DNA code $\mathcal{D} = (n, M, d, w)_4$ could be viewed as a linear code over the finite field F_4 of block length n . If for every codeword $c = (c_0, \dots, c_{n-1}) \in \mathcal{D}$, $c^* = (c_{n-1}, c_{n-2}, \dots, c_0)$ which is cyclic shift of c also belongs to $\mathcal{D}$, then $\mathcal{D}$ is a cyclic code. Through a suitable choice of irreducible polynomial, $\mathcal{D}$ becomes an irreducible cyclic code. Irreducible cyclic codes have algebraic structures that are convenient for efficient error detection and correction. Naturally, the DNA codes designing problem could be reduced to the irreducible cyclic codes computing problem in F_4 . Meanwhile, since there exists a close relationship between irreducible cyclic codes and the Gauss period, hence we use some classical conclusions from algebraic theory for attaining suitable DNA codes satisfying conditions (1) and (4). For obtaining affirmative DNA codes satisfying conditions (1), (3) and (4), we also design an algorithm to work out for the required parameters.

Theorem 3. Let $\mathbb{D}$ be an irreducible cyclic code with parameter (n, M, d) , $M = q = 4^s$, $\beta = \alpha^{\lambda}$, $c_{\beta} = (T_4^q(\beta), T_4^q(\beta\theta), \dots, T_4^q(\beta\theta^{n-1})) \in F_4^n$, where $\frac{q-1}{3} \equiv l \pmod{e}$ and $\lambda = 0, 1, 2, \dots, e-1$. Then $N_{\beta}(0) = \frac{1}{4}(n + 3\eta_0)$ ($\frac{1}{4}(n + 3\eta_1)$ or $\frac{1}{4}(n - 1)$); $N_{\beta}(1) = N_{\beta}(\omega) = N_{\beta}(\omega^2) = \frac{1}{4}(n - \eta_0)$ ($\frac{1}{4}(n - \eta_1)$ or $\frac{1}{4}(n + \eta_0 - 2\eta_1)$), where η_0 and η_1 are Gauss periods specified in Theorem 2.

Proof. Since $q - 1 = 4^s - 1 = (3 + 1)^s - 1 = e \cdot n = 3 \times \frac{q-1}{3}$, therefore, 3 is a factor of $q - 1$. Now, let $l \equiv \frac{q-1}{3} \pmod{e}$. Then, by using formulas (3), (4), (5) and (6), we compute the weights $N_{\beta}(a)$ in accordance with different cases:

1. $e \neq 3$, and $\lambda \equiv 0 \pmod{e}$,

$$N_{\beta}(0) = \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + 3\eta_0)$$

$$\begin{aligned}
N_{\beta}(1) &= \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0) \\
N_{\beta}(\omega) &= \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0) \\
N_{\beta}(\omega^2) &= \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0)
\end{aligned}$$

2. $e \neq 3$ and $\lambda \equiv 1, 2, \dots, e-1 \pmod{e}$,

$$\begin{aligned}
N_{\beta}(0) &= \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + 3\eta_1) \\
N_{\beta}(1) &= \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_1) \\
N_{\beta}(\omega) &= \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_1) \\
N_{\beta}(\omega^2) &= \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_1)
\end{aligned}$$

3. $e = 3$, $l \equiv 0 \pmod{e}$ and $\lambda \equiv 0 \pmod{e}$,

$$\begin{aligned}
N_{\beta}(0) &= \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + 3\eta_0) \\
N_{\beta}(1) &= \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0) \\
N_{\beta}(\omega) &= \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0) \\
N_{\beta}(\omega^2) &= \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0)
\end{aligned}$$

4. $e = 3$, $l \equiv 0 \pmod{e}$ and $\lambda \equiv 1, 2 \pmod{e}$,

$$\begin{aligned}
N_{\beta}(0) &= \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + 3\eta_1) \\
N_{\beta}(1) &= \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_1) \\
N_{\beta}(\omega) &= \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_1) \\
N_{\beta}(\omega^2) &= \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_1)
\end{aligned}$$

5. $e = 3$, $l \equiv 1 \pmod{e}$ and $\lambda \equiv 0 \pmod{e}$,

$$\begin{aligned}
N_{\beta}(0) &= \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 + \eta_1 + \eta_2) = \frac{1}{4}(n - 1) \\
N_{\beta}(1) &= \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 - \eta_1 - \eta_2) = \frac{1}{4}(n + \eta_0 - 2\eta_1)
\end{aligned}$$

$$N_{\beta}(\omega) = \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 - \eta_1 + \eta_2) = \frac{1}{4}(n - \eta_0)$$

$$N_{\beta}(\omega^2) = \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 + \eta_1 - \eta_2) = \frac{1}{4}(n - \eta_0)$$

6. $e = 3$, $l \equiv 1 \pmod{e}$ and $\lambda \equiv 1 \pmod{e}$,

$$N_{\beta}(0) = \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 + \eta_1 + \eta_2) = \frac{1}{4}(n - 1)$$

$$N_{\beta}(1) = \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 + \eta_1 - \eta_2) = \frac{1}{4}(n - \eta_0)$$

$$N_{\beta}(\omega) = \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 - \eta_1 - \eta_2) = \frac{1}{4}(n + \eta_0 - 2\eta_1)$$

$$N_{\beta}(\omega^2) = \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 - \eta_1 + \eta_2) = \frac{1}{4}(n - \eta_0)$$

7. $e = 3$, $l \equiv 1 \pmod{e}$ and $\lambda \equiv 2 \pmod{e}$,

$$N_{\beta}(0) = \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 + \eta_1 + \eta_2) = \frac{1}{4}(n - 1)$$

$$N_{\beta}(1) = \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 - \eta_1 + \eta_2) = \frac{1}{4}(n - \eta_0)$$

$$N_{\beta}(\omega) = \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 - \eta_1 + \eta_2) = \frac{1}{4}(n - \eta_0)$$

$$N_{\beta}(\omega^2) = \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 - \eta_1 - \eta_2) = \frac{1}{4}(n + \eta_0 - 2\eta_1)$$

8. $e = 3$, $l \equiv 2 \pmod{e}$ and $\lambda \equiv 0 \pmod{e}$,

$$N_{\beta}(0) = \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 + \eta_1 + \eta_2) = \frac{1}{4}(n - 1)$$

$$N_{\beta}(1) = \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 - \eta_1 - \eta_2) = \frac{1}{4}(n + \eta_0 - 2\eta_1)$$

$$N_{\beta}(\omega) = \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 + \eta_1 - \eta_2) = \frac{1}{4}(n - \eta_0)$$

$$N_{\beta}(\omega^2) = \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 - \eta_1 + \eta_2) = \frac{1}{4}(n - \eta_0)$$

9. $e = 3$, $l \equiv 2 \pmod{e}$ and $\lambda \equiv 1 \pmod{e}$,

$$N_{\beta}(0) = \frac{1}{4}(n + \eta_{\lambda} + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 + \eta_1 + \eta_2) = \frac{1}{4}(n - 1)$$

$$N_{\beta}(1) = \frac{1}{4}(n + \eta_{\lambda} - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 + \eta_1 - \eta_2) = \frac{1}{4}(n - \eta_0)$$

$$N_{\beta}(\omega) = \frac{1}{4}(n - \eta_{\lambda} - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 - \eta_1 + \eta_2) = \frac{1}{4}(n - \eta_0)$$

$$N_{\beta}(\omega^2) = \frac{1}{4}(n - \eta_{\lambda} + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 - \eta_1 - \eta_2) = \frac{1}{4}(n + \eta_0 - 2\eta_1)$$

10. $e = 3$, $l \equiv 2 \pmod{e}$ and $\lambda \equiv 2 \pmod{e}$,

$$\begin{aligned} N_\beta(0) &= \frac{1}{4}(n + \eta_\lambda + \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 + \eta_1 + \eta_2) = \frac{1}{4}(n - 1) \\ N_\beta(1) &= \frac{1}{4}(n + \eta_\lambda - \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 - \eta_1 + \eta_2) = \frac{1}{4}(n - \eta_0) \\ N_\beta(\omega) &= \frac{1}{4}(n - \eta_\lambda - \eta_{l+\lambda} + \eta_{2l+\lambda}) = \frac{1}{4}(n + \eta_0 - \eta_1 - \eta_2) = \frac{1}{4}(n + \eta_0 - 2\eta_1) \\ N_\beta(\omega^2) &= \frac{1}{4}(n - \eta_\lambda + \eta_{l+\lambda} - \eta_{2l+\lambda}) = \frac{1}{4}(n - \eta_0 + \eta_1 - \eta_2) = \frac{1}{4}(n - \eta_0) \quad \square \end{aligned}$$

Theorem 4. Let $\mathbb{D}$ be an irreducible cyclic code with parameter (n, M, d) , $M = q = 4^s$, e a self-conjugate prime, and t the smallest positive integer such that $2^t \equiv -1 \pmod{e}$. Then, the corresponding DNA codes satisfying the conditions (1), (4) are $C_1 = (n, n, \min\{d_1, d_2\}, w_1)_4$ and $C_2 = (n, (e-1)n, \min\{d_1, d_2\}, w_2)_4$, and the corresponding formulas of parameters are presented as follows:

$$\begin{aligned} (1) \quad l &\equiv 0 \pmod{e}, \\ w_1 &= \frac{n}{2} - \frac{1}{2e}[-1 + (e-1)2^s(-1)^{\frac{s}{t}-1}] \\ w_2 &= \frac{n}{2} - \frac{1}{2e}[-1 + 2^s(-1)^{\frac{s}{t}}] \\ d_1 &= \frac{3}{4}n - \frac{3}{4e}[-1 + (e-1)2^s(-1)^{\frac{s}{t}-1}] \\ d_2 &= \frac{3}{4}n - \frac{3}{4e}[-1 + 2^s(-1)^{\frac{s}{t}}] \\ (2) \quad l &\equiv 1, 2 \pmod{e} \\ w_1 &= \frac{n}{2} - \frac{1}{2e}[-1 + (e-1)2^s(-1)^{\frac{s}{t}-1}] \\ w_2 &= \frac{1}{2}n - \frac{1}{2e}[-1 + 2^s(-1)^{\frac{s}{t}}] \\ d_1 &= d_2 = \frac{3}{4}n + \frac{1}{4} \end{aligned}$$

where $l \equiv \frac{q-1}{3} \pmod{e}$.

Proof. From the given condition, we have two DNA codes C_1 and C_2 , the corresponding generators are c_{α^0} and c_{α^i} ($1 \leq i \leq e-1$), respectively. Since the number of all weights in c_{α^0} are n , so the number of codewords in C_1 are n . Furthermore, C_2 is generated by c_{α^i} ($1 \leq i \leq e-1$), hence, the number of codewords in C_2 are $(e-1)n$.

When $l \equiv 0 \pmod{e}$, the corresponding parameters of C_1 are presented as follows: minimum Hamming distance $d_1 = n - N_\beta(0) = N_\beta(1) + N_\beta(\omega) + N_\beta(\omega^2) = \frac{3}{4}(n - \eta_0) = \frac{3}{4}n - \frac{3}{4e}[-1 + (e-1)2^s(-1)^{\frac{s}{t}-1}]$. GC-content $w_1 = N_\beta(\omega) + N_\beta(\omega^2) = \frac{1}{2}(n - \eta_0) = \frac{1}{2}n - \frac{1}{2e}[-1 + (e-1)2^s(-1)^{\frac{s}{t}-1}]$. Similarly, we get the corresponding parameters of C_2 : $d_2 = n - N_\beta(0) = N_\beta(1) + N_\beta(\omega) + N_\beta(\omega^2) = \frac{3}{4}(n - \eta_1) = \frac{3}{4}n - \frac{3}{4e}[-1 + 2^s(-1)^{\frac{s}{t}}]$, $w_2 = N_\beta(\omega) + N_\beta(\omega^2) = \frac{1}{2}(n - \eta_1) = \frac{1}{2}n - \frac{1}{2e}[-1 + 2^s(-1)^{\frac{s}{t}}]$.

When $l \equiv 1, 2 \pmod{e}$, $N_\beta(0) = \frac{1}{4}(n - 1)$. Hence, $d_1 = d_2 = n - N_\beta(0) = N_\beta(1) + N_\beta(\omega) + N_\beta(\omega^2) = \frac{3}{4}n + \frac{1}{4}$, $w_1 = N_\beta(\omega) + N_\beta(\omega^2) = \frac{1}{2}(n - \eta_0) = \frac{n}{2} - \frac{1}{2e}[-1 + (e-1)2^s(-1)^{\frac{s}{t}-1}]$ and $w_2 = N_\beta(\omega) + N_\beta(\omega^2) = \frac{1}{2}(n - \eta_1) = \frac{1}{2}n - \frac{1}{2e}[-1 + 2^s(-1)^{\frac{s}{t}}]$.

Consequently, we get two DNA codes: $C_1 = (n, n, \min\{d_1, d_2\}, w_1)_4$ and $C_2 = (n, (e - 1)n, \min\{d_1, d_2\}, w_2)_4$. $\square$

Here we present some examples for explaining the concrete applications of [Theorem 4](#).

Example 1. $s = 3, e = 3$

Since $4^3 - 1 = e \cdot n = 3 \times 21$, $2^1 \equiv -1 \pmod{3}$, $l \equiv 0 \pmod{3}$, therefore, we utilize [Theorem 3](#) and [Theorem 4](#) for constructing suitable DNA codes. Let χ be the multiplicative character of F_{4^3} having order 3. From [Lemma 1](#) and [Theorem 1](#), we have $G(\varepsilon) = -1$, $G(\chi) = G(\chi^2) = 8$, $\eta_0 = \frac{1}{3}(-1+16) = 5$, $\eta_1 = \eta_2 = \frac{1}{3}(-1-8) = -3$, $d_1 = \frac{3}{4}(21-5) = 12$, $d_2 = \frac{3}{4}(21+3) = 18$, $w_1 = \frac{1}{2}(21-5) = 8$, $w_2 = \frac{1}{2}(21+3) = 12$. Consequently, two DNA codes are $(21, 21, 12, 8)_4$ and $(21, 42, 12, 12)_4$.

Concretely, let $f(x) = x^3 + \omega x^2 + \omega x + \omega$ be the primitive polynomial of F_4 and $F_{4^3}^* = \langle \alpha \rangle$, $\alpha^3 = \omega \alpha^2 + \omega \alpha + \omega$, then we have,

$$\begin{aligned} 1 &= (1, 0, 0), \alpha = (0, 1, 0), \alpha^2 = (0, 0, 1), \alpha^3 = (\omega, \omega, \omega), \alpha^4 = (\omega^2, 1, 1), \\ \alpha^5 &= (\omega, 1, \omega^2), \alpha^6 = (1, \omega^2, 0), \alpha^7 = (0, 1, \omega^2), \alpha^8 = (1, 1, 0), \alpha^9 = (0, 1, 1) \\ \alpha^{10} &= (\omega, \omega, \omega^2), \alpha^{11} = (1, \omega^2, \omega^2), \alpha^{12} = (1, 0, \omega), \alpha^{13} = (\omega^2, \omega, \omega^2), \alpha^{14} = (1, \omega, \omega^2) \\ \alpha^{15} &= (1, 0, \omega^2), \alpha^{16} = (1, 0, 1), \alpha^{17} = (\omega, \omega^2, \omega), \alpha^{18} = (\omega^2, 1, 0), \alpha^{19} = (0, \omega^2, 1) \\ \alpha^{20} &= (\omega, \omega, 1), \alpha^{21} = (\omega, 0, 0), \alpha^{22} = (0, \omega, 0), \alpha^{23} = (0, 0, \omega), \alpha^{24} = (\omega^2, \omega^2, \omega^2) \\ \alpha^{25} &= (1, \omega, \omega), \alpha^{26} = (\omega^2, \omega, 1), \alpha^{27} = (\omega, 1, 0), \alpha^{28} = (0, \omega, 1), \alpha^{29} = (\omega, \omega, 0) \\ \alpha^{30} &= (0, \omega, \omega), \alpha^{31} = (\omega^2, \omega^2, 1), \alpha^{32} = (\omega, 1, 1), \alpha^{33} = (\omega, 0, \omega^2), \alpha^{34} = (1, \omega^2, 1) \\ \alpha^{35} &= (\omega, \omega^2, 1), \alpha^{36} = (\omega, 0, 1), \alpha^{37} = (\omega, 0, \omega), \alpha^{38} = (\omega^2, 1, \omega^2), \alpha^{39} = (1, \omega, 0) \\ \alpha^{40} &= (0, 1, \omega), \alpha^{41} = (\omega^2, \omega^2, \omega), \alpha^{42} = (\omega^2, 0, 0), \alpha^{43} = (0, \omega^2, 0), \alpha^{44} = (0, 0, \omega^2) \\ \alpha^{45} &= (1, 1, 1), \alpha^{46} = (\omega, \omega^2, \omega^2), \alpha^{47} = (1, \omega^2, \omega), \alpha^{48} = (\omega^2, \omega, 0), \alpha^{49} = (0, \omega^2, \omega) \\ \alpha^{50} &= (\omega^2, \omega^2, 0), \alpha^{51} = (0, \omega^2, \omega^2), \alpha^{52} = (1, 1, \omega), \alpha^{53} = (\omega^2, \omega, \omega), \alpha^{54} = (\omega^2, 0, 1) \\ \alpha^{55} &= (\omega, 1, \omega), \alpha^{56} = (\omega^2, 1, \omega), \alpha^{57} = (\omega^2, 0, \omega), \alpha^{58} = (\omega^2, 0, \omega^2), \alpha^{59} = (1, \omega, 1) \\ \alpha^{60} &= (\omega, \omega^2, 0), \alpha^{61} = (0, \omega, \omega^2), \alpha^{62} = (1, 1, \omega^2). \end{aligned}$$

Let $\theta = \alpha^e = \alpha^3$, $c_\beta = (T_4^{4^3}(\beta), T_4^{4^3}(\beta\theta), T_4^{4^3}(\beta\theta^2), \dots, T_4^{4^3}(\beta\theta^{20})), \beta \in F_{4^3}^*$ and $\beta = \alpha^\lambda$, where $\lambda = 0, 1, 2$, then we have

$$\begin{aligned} c_{\alpha^0} &= (T_4^{4^3}(1), T_4^{4^3}(\alpha^3), \dots, T_4^{4^3}(\alpha^{60})) \\ c_{\alpha^1} &= (T_4^{4^3}(\alpha), T_4^{4^3}(\alpha^4), \dots, T_4^{4^3}(\alpha^{61})) \\ c_{\alpha^2} &= (T_4^{4^3}(\alpha^2), T_4^{4^3}(\alpha^5), \dots, T_4^{4^3}(\alpha^{62})) \end{aligned}$$

By computing, we get

$$\begin{aligned}c_{\alpha^0} &= (T, A, A, T, A, C, T, G, A, A, G, A, T, G, C, A, A, C, A, G, C) \\c_{\alpha^1} &= (G, G, A, C, G, G, G, C, C, A, T, C, C, C, T, T, A, G, T, T,) \\c_{\alpha^2} &= (C, G, C, G, A, G, G, T, C, T, C, A, C, C, G, T, G, T, A, T, T,)\end{aligned}$$

Also, c_{α^0} generates $C_2 = (21, 21, 12, 8)_4$ and $c_{\alpha^1}, c_{\alpha^2}$ generates $C_1 = (21, 42, 12, 12)_4$.

Example 2. $s = 4, e = 5$

Since $4^4 - 1 = e \cdot n = 5 \times 51$, $2^2 \equiv -1 \pmod{5}$, $l \equiv 0 \pmod{5}$, therefore, we utilize Theorem 3 and Theorem 4 for constructing suitable DNA codes. Let χ be the multiplicative character of F_{4^4} having order 3. From Lemma 1 and Theorem 1, we have $G(\varepsilon) = -1$, $G(\chi) = G(\chi^2) = -2^4 = -16$, $\eta_0 = \frac{1}{5}(-1 - 4 \times 16) = -13$, $\eta_1 = \eta_2 = \frac{1}{5}(-1 + 16) = 3$, $d_1 = \frac{3}{4}(51 + 13) = 48$, $d_2 = \frac{3}{4}(51 - 3) = 36$, $w_1 = \frac{1}{2}(51 + 13) = 32$, $w_2 = \frac{1}{2}(51 - 3) = 24$. Consequently, two DNA codes are $(51, 51, 36, 32)_4$ and $(51, 204, 36, 24)_4$.

Let $f(x) = x^4 + x^3 + x^2 + \omega x + \omega$ be the primitive polynomial of F_4 , $F_{4^4}^* = \langle \alpha \rangle$, $\alpha^4 = \alpha^3 + \alpha^2 + \omega \alpha + \omega$, $\theta = \alpha^e = \alpha^5$, $c_\beta = (T_4^{4^4}(\beta), T_4^{4^4}(\beta\theta), T_4^{4^4}(\beta\theta^2), \dots, T_4^{4^4}(\beta\theta^{5^4}))$, $\beta \in F_{4^4}^*$ and $\beta = \alpha^\lambda$, where $\lambda = 0, 1, 2, 3, 4$, then we have

$$\begin{aligned}c_{\alpha^0} &= (A, C, G, G, C, C, C, G, C, T, G, T, G, C, C, T, C, A, T, C, C, T, T, T, C, C, \\&\quad C, C, G, C, T, T, G, T, A, G, T, T, G, G, G, T, T, C, T, C, T, G, G, C, G) \\c_{\alpha^1} &= (T, C, G, T, G, G, A, A, C, C, A, T, G, A, A, T, G, G, T, C, G, C, C, A, A, T, \\&\quad T, A, G, C, A, A, G, C, C, G, T, C, T, T, A, A, G, G, A, C, T, A, A, C, T) \\c_{\alpha^2} &= (T, T, G, A, C, C, T, G, C, A, C, A, A, C, A, G, G, G, G, C, A, T, T, G, C, T, \\&\quad A, T, A, A, T, A, C, C, C, C, T, A, G, G, C, T, G, A, G, A, A, G, A, T, T) \\c_{\alpha^3} &= (G, T, C, A, G, T, A, T, C, G, G, A, T, A, G, A, G, C, G, T, A, C, G, A, G, T, \\&\quad C, C, A, G, A, C, A, C, T, C, G, A, T, C, A, C, G, T, T, A, C, A, T, A, T) \\c_{\alpha^4} &= (T, A, T, T, C, A, A, A, G, T, G, A, T, G, C, G, G, G, A, G, G, T, A, A, A, C, \\&\quad G, C, A, G, C, T, C, C, C, A, C, C, G, A, A, A, T, C, T, A, C, T, G, T, T)\end{aligned}$$

Also, c_{α^3} generates $C_1 = (51, 51, 36, 32)_4$ and $c_{\alpha^0}, c_{\alpha^1}, c_{\alpha^2}, c_{\alpha^4}$ generates $C_2 = (51, 204, 36, 24)_4$.

Now, we take the advantage of a table displaying the parameters of corresponding DNA codes with length $n < 1000$.

As described in Table 1, we construct new DNA codes with larger number of words and better GC-content. Most of these DNA codes achieve $\lfloor \frac{n}{2} \rfloor$ approximately, which is viewed as the idealist value of DNA codes [21]. For example, in our codes, when $n = 21$, $d = 12$, $M = 42$ and $w = 12 \approx \lfloor \frac{21}{2} \rfloor$.

Table 1

DNA codes satisfying conditions (1) and (4).

n	$M(C_1/C_2)$	$d(C_1/C_2)$	$w(C_1/C_2)$	e
5	5/10	4/4	4/2	3
21	21/42	12/12	8/12	3
51	51/204	36/36	32/24	5
85	85/170	64/64	48/40	3
93	93/930	48/48	32/48	11
315	315/3780	192/192	128/160	13
819	819/3276	576/576	384/416	5

Algorithm 1 Computation of minimum Hamming distances.**Require:** Positive integers s , e and n with $4^s - 1 = e \cdot n$.**Ensure:** DNA codes C_3 and C_4 satisfying conditions (1), (3) and (4), as well as the corresponding minimum Hamming distances d_3 and d_4 .

```

1:  $K \leftarrow GF(2, 2)$ ;  $F \leftarrow GF(2, 2s)$ 
2: Sample an irreducible  $s$ -degree polynomial over  $K$ , choose one root  $\beta$  as generator of  $F$ 
3: for  $i$  from 1 to  $4^s - 1$  do
4:    $a_i \leftarrow \beta^{i-1}$ ;  $b_i \leftarrow T_{F/K}(a_i)$ 
5:    $c_{i,j} \leftarrow b_{i+e(j-1)}$  ( $1 \leq i \leq e, 1 \leq j \leq n$ )
6:   Build codeword  $c_{\alpha^{i-1}} = [c_{i-1,1}, \dots, c_{i-1,n}]$  ( $1 \leq i \leq e$ ) and compute  $N_\beta(a)$  for all  $a \in F_4$  according
   to Theorem 3
7: end for
8: Group  $c_{\alpha^{i-1}}$  ( $i = 1, \dots, e$ ) into  $C_1, C_2$  and compute  $d_1, d_2$  according to Theorem 4
9: Let  $C_3 \leftarrow C_1 \cup \overline{C_1}^{-1}$ ;
10:  $d_3 \leftarrow d_1$ ;
11: for  $u \in C_1$  and  $v \in \overline{C_1}^{-1}$  do
12:   if  $d_3 > H(u, v)$  then
13:      $d_3 \leftarrow H(u, v)$ ;
14:   end if
15: end for
16: Let  $C_4 \leftarrow C_2 \cup \overline{C_2}^{-1}$ ;
17:  $d_4 \leftarrow d_2$ ;
18: for  $u \in C_2$  and  $v \in \overline{C_2}^{-1}$  do
19:   if  $d_4 > H(u, v)$  then
20:      $d_4 \leftarrow H(u, v)$ ;
21:   end if
22: end for
23: return  $C_3, C_4$  and  $d_3, d_4$ 

```

Next, we make further efforts for constructing valuable DNA codes satisfying conditions (1), (3) and (4). Our main technique is a post-computation algorithm. First, we construct new DNA codes $C_3 = C_1 \cup \overline{C_1}^{-1}$ and $C_4 = C_2 \cup \overline{C_2}^{-1}$, where $\overline{C_1}^{-1}$ and $\overline{C_2}^{-1}$ are the corresponding reverse Watson–Crick complements of C_1 and C_2 , respectively. Then, we combine the generators of C_1 (C_2) with the generators of $\overline{C_1}^{-1}$ ($\overline{C_2}^{-1}$) for attaining the corresponding generators of C_3 (C_4). Consequently, we obtain the minimum Hamming distance and other parameters of C_3 and C_4 .

Remark 1 (HD, GC, RC). Let $C_1 = (n, n, \min\{d_1, d_2\}, w_1)_4$ and $C_2 = (n, (e - 1)n, \min\{d_1, d_2\}, w_2)_4$ are DNA codes as Theorem 4. Then, the corresponding DNA codes satisfying conditions (1), (3) and (4) are $C_3 = (n, 2n, d_3, w_1)_4$ and $C_4 = (n, 2(e - 1)n, d_4, w_2)_4$, where d_3, d_4 are computed by post-computation algorithm given above.

For example, the generators of new DNA codes derived from Example 2 are presented as follows (see [Appendix A](#) for more details):

$$\begin{aligned}
 c_{\alpha^0} &= (A, C, G, G, C, C, C, G, C, T, G, T, G, C, C, T, C, A, T, C, C, T, T, T, C, C, C, C, G, \\
 &\quad C, T, T, G, T, A, G, T, T, G, G, G, T, T, C, T, C, T, G, G, C, G) \\
 \bar{c}_{\alpha^0}^{-1} &= (C, G, C, C, A, G, A, G, A, A, C, C, C, A, A, C, T, A, C, A, A, G, C, G, C, G, G, A, A, \\
 &\quad A, G, G, A, T, G, A, G, G, C, A, C, A, C, C, G, G, G, C, C, G, T) \\
 c_{\alpha^1} &= (T, C, G, T, G, G, A, A, C, C, A, T, G, A, A, T, G, G, T, C, G, C, C, A, A, T, T, A, G, \\
 &\quad C, A, A, G, C, C, G, T, C, T, T, A, A, G, G, A, C, T, A, A, C, T) \\
 \bar{c}_{\alpha^1}^{-1} &= (A, G, T, T, A, G, T, C, C, T, T, A, A, G, A, C, G, G, C, T, T, G, C, T, A, A, T, T, G, \\
 &\quad G, C, G, A, C, C, A, T, T, C, A, T, G, G, T, T, C, C, A, C, G, A) \\
 c_{\alpha^2} &= (T, T, G, A, C, C, T, G, C, A, C, A, A, C, A, G, G, G, G, C, A, T, T, G, C, T, A, T, A, \\
 &\quad A, T, A, C, C, C, C, T, A, G, G, C, T, G, A, G, A, A, G, A, T, T) \\
 \bar{c}_{\alpha^2}^{-1} &= (A, A, T, C, T, T, C, T, C, A, G, C, C, T, A, G, G, G, G, T, A, T, T, A, T, A, G, C, A, \\
 &\quad A, T, G, C, C, C, C, T, G, T, T, G, T, G, C, A, G, G, T, C, A, A) \\
 c_{\alpha^3} &= (G, T, C, A, G, T, A, T, C, G, G, A, T, A, G, A, G, C, G, T, A, C, G, A, G, T, C, C, A, \\
 &\quad G, A, C, A, C, T, C, G, A, T, C, A, C, G, T, T, A, C, A, T, A, T) \\
 \bar{c}_{\alpha^3}^{-1} &= (A, T, A, T, G, T, A, A, C, G, T, G, A, T, C, G, A, G, T, G, T, C, T, G, G, A, C, T, C, \\
 &\quad G, T, A, C, G, C, T, C, T, A, T, C, C, G, A, T, A, C, T, G, A, C) \\
 c_{\alpha^4} &= (T, A, T, T, C, A, A, A, G, T, G, A, T, G, C, G, G, G, A, G, G, T, A, A, A, C, G, C, A, \\
 &\quad G, C, T, C, C, C, A, C, C, G, A, A, A, T, C, T, A, C, T, G, T, T) \\
 \bar{c}_{\alpha^4}^{-1} &= (A, A, C, A, G, T, A, G, A, T, T, T, C, G, G, T, G, G, G, A, G, C, T, G, C, G, T, T, T, \\
 &\quad A, C, C, T, C, C, C, G, C, A, T, C, A, C, T, T, T, G, A, A, T, A)
 \end{aligned}$$

Therefore, $c_{\alpha^0}, \bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (51, 102, 36, 32)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, c_{\alpha^2}, \bar{c}_{\alpha^2}^{-1}, c_{\alpha^3}, \bar{c}_{\alpha^3}^{-1}, c_{\alpha^4}, \bar{c}_{\alpha^4}^{-1}$ generates $C_4 = (51, 408, 31, 24)_4$.

Similarly, we also use a table displaying the parameters of corresponding DNA codes satisfying conditions (1), (3) and (4) with length $n < 1000$.

By comparing [Table 1](#) and [Table 2](#), we observe that the minimum Hamming distances become slightly smaller to cater to the condition (3).

4. Conclusion

We constructed a new type of DNA codes with favourable properties satisfying conditions (1), (3) and (4). Being different from traditional DNA codes designing, our method

Table 2

DNA codes satisfying conditions (1), (3) and (4).

n	$M(C_3/C_4)$	$d(C_3/C_4)$	$w(C_3/C_4)$	e
5	10/20	4/4	4/2	3
21	42/84	12/12	8/12	3
51	102/408	36/31	32/24	5
85	170/340	64/53	48/40	3
93	186/1860	48/48	32/48	11
315	630/7560	192/192	128/160	13
819	1638/ 6552	573/573	384/416	5

is based on classical algebraic number theory. We first take the advantage of the relationship between irreducible cyclic codes and Gauss period for computing the weight formulas of irreducible cyclic codes. Then, we continue for producing the formulas of corresponding DNA codes using the basic theory in irreducible cyclic codes. Consequently, we use a post-computation algorithm for providing the parameters of suitable DNA codes. Also, our method is universal. One can take the advantage of this technique for constructing other new valuable DNA codes.

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Appendix A

- When $n = 5$, $e = 3$, let $f(x) = x^2 + x + \omega$ be the primitive polynomial of F_4 , then c_{α^0} generates $C_1 = (5, 5, 4, 4)_4$, $c_{\alpha^1}, c_{\alpha^2}$ generate $C_2 = (5, 10, 4, 2)_4$, $c_{\alpha^0}, \bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (5, 10, 4, 4)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, c_{\alpha^2}, \bar{c}_{\alpha^2}^{-1}$ generate $C_4 = (5, 20, 4, 2)_4$.
- When $n = 21$, $e = 3$, let $f(x) = x^3 + \omega x^2 + \omega x + \omega$ be the primitive polynomial of F_4 , then c_{α^0} generates $C_1 = (21, 21, 12, 8)_4$, $c_{\alpha^1}, c_{\alpha^2}$ generate $C_2 = (21, 42, 12, 12)_4$, $c_{\alpha^0}, \bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (21, 42, 12, 8)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, c_{\alpha^2}, \bar{c}_{\alpha^2}^{-1}$ generate $C_4 = (21, 84, 12, 12)_4$.
- When $n = 51$, $e = 5$, let $f(x) = x^4 + x^3 + x^2 + \omega x + \omega$ be the primitive polynomial of F_4 , then c_{α^0} generates $C_1 = (51, 51, 36, 32)_4$, $c_{\alpha^1}, c_{\alpha^2}, c_{\alpha^3}, c_{\alpha^4}$ generate $C_2 = (51, 204, 36, 24)_4$, $c_{\alpha^0}, \bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (51, 102, 36, 32)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, c_{\alpha^2}, \bar{c}_{\alpha^2}^{-1}, c_{\alpha^3}, \bar{c}_{\alpha^3}^{-1}, c_{\alpha^4}, \bar{c}_{\alpha^4}^{-1}$ generate $C_4 = (51, 408, 31, 24)_4$.
- When $n = 85$, $e = 3$, let $f(x) = x^4 + x^3 + x^2 + \omega x + \omega$ be the primitive polynomial of F_4 , then c_{α^0} generates $C_1 = (85, 85, 64, 48)_4$, $c_{\alpha^1}, c_{\alpha^2}$ generate $C_2 = (85, 170, 64, 40)_4$, $c_{\alpha^0}, \bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (85, 170, 64, 48)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, c_{\alpha^2}, \bar{c}_{\alpha^2}^{-1}$ generate $C_4 = (85, 340, 53, 40)_4$.
- When $n = 93$, $e = 11$, let $f(x) = x^5 + \omega x^4 + \omega x^3 + \omega x^2 + \omega x + \omega$ be the primitive polynomial of F_4 , then c_{α^0} generates $C_1 = (93, 93, 48, 32)_4$, $c_{\alpha^1}, c_{\alpha^2}, \dots, c_{\alpha^{10}}$

- generate $C_2 = (93, 930, 48, 48)_4$, c_{α^0} , $\bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (93, 186, 48, 32)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, \dots, c_{\alpha^{10}}, \bar{c}_{\alpha^{10}}^{-1}$ generate $C_4 = (93, 1860, 48, 48)_4$.
6. When $n = 315$, $e = 13$, let $f(x) = x^6 + \omega^2 x^2 + \omega x + \omega$ be the primitive polynomial of F_4 , then c_{α^0} generates $C_1 = (315, 315, 192, 128)_4$, $c_{\alpha^1}, c_{\alpha^2}, \dots, c_{\alpha^{12}}$ generate $C_2 = (315, 3780, 192, 160)_4$, c_{α^0} , $\bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (315, 630, 192, 128)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, \dots, c_{\alpha^{12}}, \bar{c}_{\alpha^{12}}^{-1}$ generate $C_4 = (315, 7560, 192, 160)_4$.
7. When $n = 819$, $e = 5$, let $f(x) = x^6 + \omega^2 x^2 + \omega x + \omega$ be the primitive polynomial of F_4 , then c_{α^0} generates $C_1 = (819, 819, 576, 384)_4$, $c_{\alpha^1}, \dots, c_{\alpha^4}$ generate $C_2 = (819, 3276, 576, 416)_4$, c_{α^0} , $\bar{c}_{\alpha^0}^{-1}$ generate $C_3 = (819, 1638, 573, 384)_4$ and $c_{\alpha^1}, \bar{c}_{\alpha^1}^{-1}, \dots, c_{\alpha^4}, \bar{c}_{\alpha^4}^{-1}$ generate $C_4 = (819, 6552, 573, 416)_4$.

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